

AKASH YADAV

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RESEARCH INTERESTS

AI4Science, Scientific Foundation Models, Uncertainty Quantification, Probabilistic Machine Learning, Reduced-order Modeling, and Computational Mechanics

EDUCATION

Doctor of Philosophy, Civil Engineering (GPA: 4.0/4.0) 08/2023 - 05/2027 (expected)
Thesis: Quantify and Reduce Model-error in Physics-based and Learned Models via Stochastic Representations
University of Houston (UH) - Houston, USA

Master of Technology (Research), Civil Engineering (GPA: 8.3/10) 10/2020 - 06/2023
Thesis: SHM Accounting for Thermal Variability and Damage Using Approximate Bayesian Computation
Indian Institute of Science (IISc) - Bangalore, India

Bachelor of Technology, Civil Engineering (GPA: 8.75/10) 07/2014 - 05/2018
Indian Institute of Technology (IIT) - Roorkee, India

SKILLS

ML/UQ	Transformers, LoRA, PINNs, CNNs, GANs, Gaussian Processes, BO, MCMC
Programming	Python, PyTorch, GPyTorch, BoTorch, NumPy, SciPy, MATLAB, Julia
HPC/CAE	Distributed training, SLURM, Git, Abaqus, LS-DYNA, FEniCS, Gmsh

RESEARCH EXPERIENCE

Graduate Research Assistant - UH (Advisor: Dr. Ruda Zhang) 08/2023 - Present

- **Uncertainty-aware foundation models:** Developed inference-time stochastic attention that equips pretrained or fine-tuned scientific foundation models (SciFMs) with calibrated uncertainty without re-training, validated on weather, time-series, and PDE SciFMs while preserving predictive accuracy.
- **Stochastic subspaces for model error:** Randomized the projection subspace of reduced-order models via probabilistic PCA and the Bootstrap to characterize the error of high-fidelity simulations and their reduced models, with 95% prediction intervals up to 45× tighter than the state of the art.
- **Bayesian optimization under uncertainty:** Built a workflow to tune hyperparameters in stochastic models, reducing data requirements and accelerating tuning by 40× relative to Monte Carlo-based optimization and 15× relative to conventional Gaussian-process Bayesian optimization.
- **Model error correction:** Developed a stochastic reduced-order model (SROM) framework whose low-dimensional correction terms absorb systematic discrepancy in computational mechanics, improving predictive uncertainty under limited data.

Graduate Research Assistant - IISc (Advisor: Dr. Ananth Ramaswamy) 10/2020 - 06/2023

- Developed a method based on Approximate Bayesian Computation (ABC) for damage detection under varying temperature conditions, and extended it to incorporate damage-induced nonlinearity.

SELECTED PUBLICATIONS

1. **Yadav, A., Adebisi, T.A., & Zhang, R.** (2026). "Calibrating Scientific Foundation Models with Inference-Time Stochastic Attention." *Advances in Neural Information Processing Systems (NeurIPS 2026)*. [arXiv:2604.19530](#).
2. **Yadav, A. & Zhang, R.** (2026). "Bayesian Optimization under Uncertainty for Training a Scale Parameter in Stochastic Models." *ASCE-ASME J. Risk Uncertainty Eng. Syst., Part A: Civ. Eng.*, 12(4), 04026050. [doi:10.1061/AJRUA6.RUENG-1854](#).
3. **Yadav, A. & Zhang, R.** (2026). "Nonparametric Stochastic Subspaces via the Bootstrap Method for Characterizing Model Error." *ASCE-ASME J. Risk Uncertainty Eng. Syst., Part A: Civ. Eng.*, 12(4), 04026061. [doi:10.1061/AJRUA6.RUENG-1948](#).
4. **Yadav, A. & Zhang, R.** (2026). "Stochastic Subspace via Probabilistic Principal Component Analysis for Characterizing Model Error." *Computational Mechanics*, 77, 1141–1156. [doi:10.1007/s00466-025-02701-6](#).

PROJECTS

Trustworthy AI for CO₂ Storage Monitoring

08/2026 - Present

Supported by the UH Chevron Energy Graduate Fellows Award

- Developing AI surrogates that forecast CO₂ plume migration and pressure with calibrated uncertainty.
- Designing Bayesian history matching that updates both the geology and the surrogate's own error model.

IC-SHM 2026: Structural Damage Diagnosis from Images and Text

06/2026 - 09/2026

- Led a three-member team (UH, IISc) that built an expert committee in which a fine-tuned vision-language model (Qwen3-VL-8B) and two image classifiers vote on the damage types, reaching a micro-F1 of 0.98.
- Showed that fine-tuning mainly teaches the model the annotators' vocabulary, not new visual skills.

SELECTED CONFERENCE PRESENTATIONS

1. **Yadav, A.**, Adebisi, T.A., & Zhang, R. "Calibrated Uncertainty for Fine-tuned Scientific Foundation Models via Stochastic Attention." *WCCM/ECCOMAS 2026*, Munich, Germany.
2. **Yadav, A.** & Zhang, R. "Stochastic Reduced-order Modeling for Model Error Characterization and Correction." *USNCCM 2025*, Chicago, USA.
3. **Yadav, A.** & Zhang, R. "Stochastic Subspace via Bootstrap for Model-form Uncertainty." *CASML 2024*, IISc Bangalore, India.
4. **Yadav, A.** & Zhang, R. "Stochastic Subspace via Probabilistic Principal Component Analysis for Model-form Uncertainty." *WCCM/PANACM 2024*, Vancouver, Canada.
5. **Yadav, A.** & Ramaswamy, A. "Structural Health Monitoring of Steel Truss Bridges Subjected to Environmental Variability." *ICCMS 2022*, IIT Indore, India. Published as a [Springer book chapter](#).

HONORS AND AWARDS

Chevron Energy Graduate Fellows Award, UH

08/2026 - 05/2027

Jimmie A. Schindewolf Academic Scholarship, UH

08/2024, 2026

Student Travel Award for WCCM/ECCOMAS 2026

03/2026

Presidential Fellowship, UH

08/2023 - 05/2025

Finalist, USACM UQ Student Paper Competition, WCCM/PANACM 2024

07/2024

Cullen Fellowship Travel Grant, UH

05/2024

Top 0.35% in IIT-JEE (out of 1.4 million candidates)

06/2014

TEACHING EXPERIENCE

Teaching Assistant, Foundation Engineering (CIVE 4369), UH

01/2026 - 05/2026

Recitation Instructor, Mechanics-I Statics (ENGR 2301), UH

08/2024 - 12/2024

INDUSTRY EXPERIENCE

Senior Project Engineer - Indian Oil Corporation Limited

07/2018 - 09/2020

Oversaw civil execution for an energy-efficient green building and process-unit expansion; coordinated contractors, ensured code compliance, and managed schedules across multiple work fronts.

RELEVANT COURSEWORK

Deep Learning, Mathematics for AI, Learning with Data, Data-Driven Engineering, Bayesian Statistics, Finite Element Method, Numerical Methods, Optimization Methods.

EXTRACURRICULAR ACTIVITIES

Secretary/Joint Secretary, Taekwondo, Institute Sports Council, IIT Roorkee

2016 - 2018

First Dan Black Belt in Taekwondo by World Taekwondo Federation

08/2017

Represented State Uttarakhand in **National Taekwondo Championship**

12/2017

ACADEMIC AND PROFESSIONAL AFFILIATIONS

United States Association for Computational Mechanics (USACM), Graduate Student Member, Technical Thrust Area in Uncertainty Quantification and Probabilistic Modeling.

Society for Industrial and Applied Mathematics (SIAM), Graduate Student Member.